Significance of Selected Research

Wentao Zhang · ORCID 0000-0003-1438-9056 · Google Scholar
Nomination for the *Electronics* Young Investigator Award 2026

My work develops **physics-informed AI and validated digital twins for power-electronic and electromechanical systems**. The four works below evidence one capability, namely turning slow, high-fidelity physics into fast, accurate, and differentiable models usable in design and real-time operation. The name in **bold** is the applicant; identifiers refer to the accompanying List of Publications.

1 — Physics-informed AI surrogate (methodological centre)

W. Zhang, Z. Wu, W. Hua, "A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs," Proc. ICEM, 2026 (accepted). [C1]

This is the methodological core of my digital-twin research and the work most directly aligned with the journal's Artificial Intelligence and Power Electronics scope. It develops an **implicit-physics deep-learning surrogate**, with a Fourier coordinate-encoding layer, that maps design parameters to the **full magnetic vector potential field** rather than to scalar outputs alone, reaching $R^2 = 0.9996$ for average torque and 0.9730 for iron loss on unseen designs, with **more than 94% inference speed-up** over finite-element analysis. Implemented in PyTorch, it is differentiable end-to-end, enabling sensitivity analysis and gradient-based design. It is the layer that turns a physics model into a **real-time, differentiable digital twin**. *Contribution:* as sole lead author I designed the network architecture and encoding scheme, generated the training data with my in-house finite-element code, trained and validated the model, and wrote the paper.

2 — Validated multiphysics digital twin

W. Zhang, Z. Wu, W. Hua, "High-Fidelity Lumped Parameter Thermal Model for Oil-Cooling Hairpin Winding PM Machines in EVs," IEEE Trans. Transport. Electrific., vol. 12, no. 2, pp. 2978–2991, 2026. [J10]

The clearest demonstration in my record of a **validated, real-time-capable digital twin**. It develops a conductor-resolved thermal model with fluid-solid coupled cooling that captures transient coolant mass-transfer, validated against dynamometer measurements across the full operating map, returning temperature fields in seconds where an equivalent simulation needs many CPU-hours. *Contribution:* as first author I formulated the coupled model, implemented it, designed and ran the experimental validation, and wrote the manuscript.

3 — Manufacturing-aware power-density design (Prize Paper)

W. Zhang, Z. Wu, W. Hua, "Design of n-Layer Flat Wire Winding in Permanent Magnet Machine for Electric Vehicles," IEEE Trans. Transport. Electrific., vol. 11, no. 3, pp. 7798–7810, 2025. (2025 IEEE TTE Prize Paper Award, 2nd Prize.) [J5]

Recognised by an IEEE Transactions Prize Paper Award, this work generalises high power-density flat-wire (hairpin) winding design from even to **arbitrary layer numbers** and folds conductor layout, mechanical and tolerance-aware design, and manufacturing constraints into one methodology, connecting design intent to the device that is actually built. *Contribution:* as first author I conceived the generalised methodology, carried out the analysis and optimisation, prepared the design for manufacture, and led the writing.

4 — High-speed rotor mechanical and acoustic design

W. Zhang, Z. Wu, W. Hua, "Design of Notching and Mechanical Stress Relieving Grooves for High-Speed PM Machine in EVs," IEEE Trans. Ind. Appl., early access, 2026. [J11]

Rotor surface notching suppresses torque ripple and the key-order radial force responsible for acoustic noise, while the resulting lamination **stress concentration** is resolved by stress-relieving grooves, cutting maximum stress from 387 MPa to 369 MPa. A high-speed prototype was **built and tested**, including a **22,500 rpm mechanical overspeed test** and acoustic measurement, with results matching prediction. *Contribution:* as first author I formulated the notching analysis, designed the grooves, ran the coupled structural and electromagnetic studies, led the build and testing, and wrote the manuscript.

Full-text copies of [C1], [J10], [J5], and [J11] are provided in the supporting documents.